MoKE: A Mixture of Koopman Experts Framework for Modeling Nonlinear Dynamical Systems

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Abstract

We present a Mixture of Koopman Experts (MoKE) framework that models nonlinear dynamical systems with multiple equilibria and complex regime transitions. Koopman operator theory supplies a linear representation of nonlinear dynamics in an infinite-dimensional observable space, but a single global Koopman operator cannot capture systems with several locally linearizable regimes. Our approach learns a set of local Koopman operators, each describing dynamics around a specific equilibrium, and a state-dependent gating network that blends these operators to create smooth regime transitions. We implement the MoKE framework and evaluate it on four canonical nonlinear systems: the Duffing oscillator, Van der Pol oscillator, Lorenz attractor, and double pendulum. Using 100,000 trajectories per system with train/validation splits, we compare MoKE against Vector Autoregression (VAR), Extended Dynamic Mode Decomposition (eDMD), and single-operator Koopman Autoencoders. Tests on 100 held-out trajectories per system show that MoKE with 3–4 experts delivers the best prediction accuracy within the training horizon (1–100 steps) and maintains zero trajectory divergence during extrapolation to 500–1000 steps. In contrast, eDMD diverges on 16% of Duffing trajectories and 15% of Double Pendulum trajectories. We also analyze Lyapunov times to explain why long-horizon extrapolation performance converges across all models for chaotic systems - a fundamental physics limit rather than a modeling flaw. Our work unifies the interpretability of Koopman theory with the flexibility of neural mixture architectures and creates a robust foundation for modeling multi-regime nonlinear dynamics.

https://github.com/pujiacheng/ESE5380_MoKE-on-non-linear-dynamical-system

Index Terms

Koopman Operator, Mixture-of-Experts, Nonlinear Dynamics, Dynamical Systems, Neural Networks, Time Series Prediction

Notation

$x_t \in \mathbb{R}^{n_x}$	State vector at time t
$z_t \in \mathbb{R}^{n_z}$	Latent (observable) vector at time t
$\dot{x}$	first derivative of x with respect to time t
$\ddot{x}$	second derivative of x with respect to time t
$f(\cdot)$	State transition map
$g(\cdot)$	Observable function
$\phi(\cdot), \psi(\cdot)$	Encoder and decoder networks
$\mathcal{K}$	Infinite-dimensional Koopman operator
K, A_f	Finite-dimensional Koopman matrix (single operator)
K_k	k-th expert Koopman operator, $k \in \{1, \ldots, M\}$
$\varphi(\cdot)$	Eigenfunction
M	Number of experts
$\pi_k(z)$	Gating weight for expert k
$\rho(K)$	Spectral radius of matrix K
$\boldsymbol{\phi}(\cdot)$	Dictionary functions in eDMD
$\mathcal{L}_{\text{rec}}, \lambda_{\text{rec}}$	Reconstruction Loss and its weight
$\mathcal{L}_{\text{pred}}, \lambda_{\text{pred}}$	Prediction Loss and its weight
$\mathcal{L}_{\text{lin}}, \lambda_{\text{lin}}$	Multi-Step Linearity Loss and its weight
$\mathcal{L}_{\text{bi}}, \lambda_{\text{bi}}$	Bidirectional Consistency Loss and its weight
$\mathcal{L}_{\text{spec}}, \lambda_{\text{spec}}$	Spectral Radius Penalty Loss and its weight
$\mathcal{L}_{\text{bal}}, \lambda_{\text{bal}}$	Load Balancing Loss and its weight
$\mathcal{L}_{\text{sparse}}, \lambda_{\text{sparse}}$	Sparsity Regularization Term and its weight
λ_{Lyap}	Lyapunov exponent (for chaotic systems)
W	Weights in neural network layers
b	bias in neural network layers
θ	All trainable parameters
H	Prediction horizon
N	Number of training samples
i	Index over samples (in loss functions)
k	Index over experts

I. Introduction

Modeling nonlinear dynamical systems remains one of the central challenges in system identification, control, and reinforcement learning. Many real-world systems such as oscillators, pendulums, and fluid flows exhibit multiple equilibria or attractor regimes with qualitatively distinct dynamics. Classical linearization methods fail to capture transitions between these regimes, motivating the development of structured, data-driven representations that preserve interpretability while offering flexibility. Koopman operator theory [1], [2] provides a powerful framework for analyzing nonlinear dynamics by lifting the state evolution into an infinite-dimensional space of observables where the dynamics evolve linearly. For a discrete-time dynamical system $x_{t+1} = f(x_t)$, the Koopman operator $\mathcal{K}$ acts on observable functions $g : \mathbb{R}^{n_x} \to \mathbb{R}$ such that:

$$(\mathcal{K}g)(x) = g(f(x)) \tag{1}$$

This elegant formulation enables the application of linear analysis tools to inherently nonlinear problems. However, practical implementations typically employ a single finite-dimensional approximation of the Koopman operator, which struggles to capture global nonlinearities or systems with multiple local equilibria. Classical Koopman approaches assume a single globally valid linear embedding - an assumption fundamentally violated by systems with multiple attractors, each characterized by distinct local dynamics. This limitation motivates the need for multiple locally linear Koopman operators combined through a differentiable gating mechanism that can smoothly transition between dynamic regimes while maintaining interpretability. To address these challenges, we propose the *Mixture of Koopman Experts (MoKE)* framework - a differentiable, data-driven architecture that blends multiple local Koopman operators through a state-dependent gating mechanism. By decomposing complex multi-regime dynamics into specialized local operators while preserving the linear structure of Koopman theory, MoKE achieves both the interpretability of classical operator-theoretic methods and the flexibility required for systems with multiple equilibria. Each expert captures dynamics within a specific attractor basin, while the gating network enables smooth, stable transitions across regime boundaries.

A. Contributions

The main contributions of this work are:

1) A novel MoKE architecture combining shared encoder-decoder networks with multiple Koopman operators and learned gating weights (see Figure 1)
2) A comprehensive training methodology using dense multi-step prediction horizons (1-100 steps)
3) Extensive empirical evaluation on four canonical nonlinear systems with 100000 training and validation trajectories each and 100 held-out test trajectories per system, demonstrating MoKE's performance superiority over baselines (see Tables V)
4) Demonstration of MoKE's stability: zero divergence across 100 test trajectories vs. 16% divergence rate for eDMD (see Tables II)

II. Related Work

A. Koopman Operator Theory and Data-Driven Approximations

The Koopman operator, introduced by Koopman [1], provides a linear representation of nonlinear dynamics in function space. Mezić [2] built the spectral analysis framework that drives most modern Koopman-based applications and enables modal decompositions for high-dimensional systems. The Koopman mode decomposition

$$g(x) = \sum_{j=1}^{\infty} v_j \varphi_j(x) \tag{2}$$

separates spatial modes v_j from temporal dynamics governed by eigenvalues λ_j. This structure allows researchers to study stability, oscillatory behavior, and long-term invariants through linear analysis tools, even when the underlying dynamics remain strongly nonlinear. Practitioners have used this decomposition for fluid flows, robotics, power systems, and climate dynamics because it yields interpretable temporal spectra and physically meaningful mode shapes. Dynamic Mode Decomposition (DMD) [3] provides a data-driven algorithm that approximates Koopman spectral properties from trajectory snapshots. DMD computes a best-fit linear operator that advances measured states and extracts eigenvalues and modes that characterize dominant coherent structures. Extended DMD (EDMD) [4] improves approximation quality by lifting data into a richer observable space using hand-crafted or kernel-induced dictionary functions. EDMD can capture more complex dynamics but depends heavily on the representational capacity of the chosen dictionary. Despite their utility, both DMD and EDMD rely on a single global operator assumption, which restricts their ability to represent systems with multiple equilibria, strong nonlinearity, or state-dependent linearization behavior.

B. Neural Koopman Methods

Deep learning approaches to Koopman operator approximation have expanded rapidly over the past decade. Lusch et al. [5] introduced deep autoencoders that learn nonlinear observable maps capable of capturing invariant subspaces more effectively than manual dictionaries. Their work highlighted the advantage of nonlinear feature learning in discovering Koopman coordinates for systems such as the Duffing and forced pendulum. Li et al. [6] proposed end-to-end differentiable Koopman networks that integrate the encoder, decoder, and linear evolution operator into a single training pipeline, enabling gradient-based learning of latent dynamics tailored to dataset-specific structure. Neural Koopman methods employ nonlinear encoder-decoder pairs (ϕ, ψ) that map trajectories into a latent space where a linear operator governs evolution as $z_{t+1} = Kz_t, z_t = \phi(x_t), \hat{x}_t = \psi(z_t)$. These models leverage expressive neural architectures, regularization schemes, and spectral constraints to stabilize training and enforce linearity in latent space. Researchers have extended these ideas to control, stochastic modeling, and Hamiltonian systems. However, single-operator neural Koopman formulations still assume a globally linearizable regime and therefore struggle with systems that exhibit multiple attractors, switching behaviors, or broad nonlinear variation.

C. Multi-Regime and Switching Models

Researchers have proposed several approaches to address multi-regime dynamics directly. Kaiser et al. [7] developed switching Koopman models that introduce discrete latent states to index different Koopman operators. This architecture can represent systems with abrupt regime changes but requires reliable segmentation of the state space and often faces identifiability challenges. Proctor et al. [8] presented piecewise-linear DMD, where separate DMD models represent different bifurcation regimes. Their method works well when practitioners can manually identify and partition regimes but does not generalize smoothly to cases where regime boundaries remain unknown or continuous. Other multi-regime approaches include hybrid systems modeling, symbolic regression of regime indicators, and clustering-driven operator banks. Although these methods increase expressive power, they typically rely on hard switching, which creates discontinuities at regime boundaries and limits extrapolation stability. They also depend on explicit partitioning strategies that degrade performance when the underlying system transitions gradually or exhibits overlapping local linearizations.

D. Mixture of Experts

The mixture of experts (MoE) framework [9] decomposes complex functions by combining specialized expert networks with a learned gating mechanism. Each expert handles a subset of the input space, while the gating network computes soft routing weights to blend expert outputs. This architecture supports modularity, interpretability, and efficient specialization—attributes that align well with systems exhibiting regime-specific structure. Advances in scalable MoE design, such as sparsely activated layers [10], have shown how soft routing can improve accuracy and computational efficiency in large-scale models while preventing experts from collapsing to identical representations. Yang et al. [11] present a comprehensive survey of MoE architectures, covering applications in vision, language modeling, reinforcement learning, and scientific computing. Their analysis highlights the ability of MoE systems to manage heterogeneous data distributions and to learn decompositions aligned with underlying generative mechanisms. Our work leverages these strengths by adapting the MoE paradigm specifically to the structure of Koopman-based dynamical system modeling, where local linear operators naturally correspond to expert components.

E. Positioning of MoKE

Table XII summarizes the landscape of existing Koopman and multi-regime modeling methods and positions MoKE within this context. Existing Koopman techniques either rely on a single operator—limiting their ability to model multiple equilibria and nonuniform nonlinearities—or use hard-switching approaches that undermine stability and smoothness near regime boundaries. MoE-based systems offer soft, differentiable transitions but have not previously been tailored to the constraints of Koopman theory. MoKE bridges these gaps by combining Koopman-theoretic structure with differentiable mixture-of-experts design. The framework learns multiple local Koopman operators that describe distinct linearized regimes and a continuous gating network that performs soft routing based on state. This combination enables MoKE to model multi-equilibrium systems, capture state-dependent transitions, and retain interpretability through operator-level decomposition. As the table illustrates, MoKE uniquely occupies the intersection of Koopman-based interpretability, multi-regime flexibility, and smooth expert transitions, thereby addressing core limitations of prior methods.

III. Problem Formulation

A. Discrete-Time Dynamical Systems

We study discrete-time nonlinear dynamical systems governed by the iterative process

$$x_{t+1} = f(x_t), \qquad x_t \in \mathbb{R}^{n_x}, \tag{3}$$

where the transition map $f : \mathbb{R}^{n_x} \to \mathbb{R}^{n_x}$ is unknown and may exhibit complex behavior. The structure of f can vary dramatically across the state space. For example, the Jacobian $Df(x)$ may differ significantly between equilibria, along limit cycles, and in regions dominated by chaotic stretching and folding dynamics. Such variation implies the existence of multiple locally valid linear approximations, each describing the system's behavior near a specific attractor or saddle manifold. The overall system evolution may involve slow drifts, fast transients, or transitions across separatrices that divide basins of attraction. When analyzing or predicting such dynamics, a global representation must reconcile these heterogeneous local behaviors while remaining robust under rollout. This requirement motivates latent representations that support piecewise-linear structure while remaining expressive enough to approximate nonlinear phenomena.

B. Koopman Operator Theory

Koopman operator theory provides a strategy for lifting nonlinear dynamics into a space where linear evolution becomes possible. For any scalar-valued observable function $g : \mathbb{R}^{n_x} \to \mathbb{R}$, the Koopman operator $\mathcal{K}$ acts as

$$(\mathcal{K}g)(x_t) = g(f(x_t)) = g(x_{t+1}). \tag{4}$$

This operator is linear despite the potential nonlinearity of f. Because $\mathcal{K}$ acts on functions rather than states, its domain is infinite dimensional. The associated spectral properties such as eigenfunctions, eigenvalues, and Koopman modes characterize invariant sets, coherent structures, and recurrence properties of the underlying dynamics. In many cases, one seeks observable functions $\{\varphi_j\}$ that approximately span an invariant subspace under $\mathcal{K}$:

$$\mathcal{K}\varphi_j = \lambda_j \varphi_j, \tag{5}$$

yielding a modal decomposition of the form

$$g(x_t) = \sum_{j=1}^{n_z} v_j \varphi_j(x_t) + r(x_t), \tag{6}$$

where r denotes the residual outside the chosen subspace. When the residual is small, the system evolves approximately linearly in the coordinates $\varphi_j(x)$. This motivates learning latent embeddings that approximate such coordinates. Because direct representation of $\mathcal{K}$ is generally infeasible, we approximate it with a finite-dimensional operator $K \in \mathbb{R}^{n_z \times n_z}$ acting on a learned latent state:

$$z_t = \phi(x_t) \tag{7}$$

$$z_{t+1} = K z_t \tag{8}$$

$$\hat{x}_{t+1} = \psi(z_{t+1}). \tag{9}$$

The encoder ϕ aims to learn coordinates that are as close as possible to Koopman eigenfunctions or their linear combinations, such that K becomes approximately linear and time-invariant. The decoder ψ reconstructs the state by inverting the learned embedding. Training such models therefore involves jointly shaping ϕ, ψ, and K to minimize prediction error while preserving approximate linearity of latent evolution.

C. Limitation of Single-Operator Models

A single Koopman operator K provides a global linear approximation of the nonlinear map f in the learned latent space. This requires the dynamics to admit a single linear model that is valid across all regions visited by the system. In general, nonlinear systems do not satisfy this condition. The local linearization

$$x_{t+1} \approx f(x^\star) + Df(x^\star)(x_t - x^\star) \tag{10}$$

depends on the point $x^\star$ at which it is computed. When the system evolves through regions with different Jacobians $Df(x)$ generally vary across the state space, and the corresponding local linearizations differ as well. For instance, systems with multiple equilibria admit distinct linearizations $Df(x_i^\star)$ at each equilibrium; systems with periodic orbits produce a time-varying Jacobian $Df(x(t))$ along the cycle, leading to a monodromy matrix that depends on the orbit; and systems with chaotic trajectories exhibit position-dependent expansion and contraction rates, which appear as spatial variation in $Df(x)$. These differences imply that no single linear operator can accurately represent the dynamics across all regions visited by the system. Because these local linear models are not identical, no single finite-dimensional linear operator K can represent all of them simultaneously. A global operator must compromise between different regions, which forces it to approximate each local linearization only coarsely. As trajectories move between regions with incompatible linearizations, prediction error accumulates and eventually dominates long-horizon rollouts. This mismatch motivates using multiple operators, each aligned with a different local linear approximation, rather than enforcing a single global operator for all regimes.

D. Mixture of Koopman Experts Formulation

We introduce the Mixture of Koopman Experts (MoKE) framework, which represents the dynamics using a set of M local Koopman operators, $K_1, K_2, \ldots, K_M \in \mathbb{R}^{n_z \times n_z}$, each intended to approximate the linearized evolution of the system within a specific region of the latent space. These operators correspond to different locally valid approximations of the Koopman operator restricted to functional subspaces associated with distinct attractors or nonlinear structures. A gating network $\pi : \mathbb{R}^{n_z} \to \Delta^{M-1}$ assigns mixture weights based on the current latent state. The output vector

$$\pi(z_t) = [\pi_1(z_t), \ldots, \pi_M(z_t)] \tag{11}$$

satisfies

$$\sum_{k=1}^{M} \pi_k(z_t) = 1, \qquad \pi_k(z_t) \geq 0. \tag{12}$$

The geometry of the simplex Δ^{M-1} ensures that the gating mechanism provides a convex combination of operators, which guarantees smooth transitions between regimes and prevents discontinuities in latent dynamics. The latent evolution rule becomes

$$z_{t+1} = \sum_{k=1}^{M} \pi_k(z_t) K_k z_t \tag{13}$$

IV. MoKE Framework Architecture

A. Overall Architecture

Figure 1 shows the MoKE architecture. The framework contains four components: a shared encoder ϕ that maps states into a latent observable space, a gating network that assigns mixture weights, a set of Koopman operators $\{K_k\}_{k=1}^{M}$ that approximate locally linear dynamics, and a shared decoder ψ that reconstructs states from latent coordinates. Each component contributes to producing a latent trajectory that evolves under a state-dependent convex combination of linear operators.

B. Network Components

1) Shared Encoder: The encoder $\phi : \mathbb{R}^{n_x} \to \mathbb{R}^{n_z}$ maps the input state into a latent coordinate system intended to approximate a finite-dimensional Koopman-invariant subspace. We implement ϕ as a feedforward network,

$$\phi(x) = W_3\, \sigma(W_2\, \sigma(W_1 x + b_1) + b_2) + b_3, \tag{14}$$

where σ denotes the ReLU activation function. In our implementation, we use a 4-layer MLP with hidden dimension 128 and LayerNorm after each linear transformation for stable training. The latent dimension is set to $n_z = 5 \times n_x$ to provide sufficient capacity for representing multiple local linearizations while keeping the latent space tractable.

2) Gating Network: The gating network computes the mixture weights that determine how the Koopman operators contribute at each time step. Given a latent state $z \in \mathbb{R}^{n_z}$, it outputs

$$\pi(z) = \mathrm{softmax}(W_g z + b_g) \in \Delta^{M-1}, \tag{15}$$

where Δ^{M-1} denotes the probability simplex. Because the softmax produces nonnegative components that sum to one, the gating mechanism yields a convex combination of expert operators, ensuring that the effective transition matrix remains a weighted average of $\{K_k\}$.

3) Koopman Operators: Each expert k maintains a trainable Koopman operator $K_k \in \mathbb{R}^{n_z \times n_z}$ which approximates the locally linear dynamics in one region of the latent space. To encourage near-identity behavior at initialization, we set

$$K_k^{(0)} = I + \epsilon A_k, \qquad A_k \sim \mathcal{N}(0,\, 0.01), \tag{16}$$

with a small scalar ϵ. We also learn inverse operators K_k^{-1} to impose bidirectional consistency constraints in training, ensuring that forward and backward evolution agree along latent trajectories.

4) Shared Decoder: The decoder $\psi : \mathbb{R}^{n_z} \to \mathbb{R}^{n_x}$ reconstructs states from latent coordinates and mirrors the encoder architecture. It provides the mapping needed to obtain state-space predictions after evolving the latent representation under the mixture of operators.

C. Forward Dynamics

Given an initial condition x_0, a single-step prediction proceeds as

$$z_0 = \phi(x_0), \tag{17}$$

$$\pi^{(0)} = \pi(z_0), \tag{18}$$

$$z_1 = \sum_{k=1}^{M} \pi_k^{(0)} K_k z_0, \tag{19}$$

$$\hat{x}_1 = \psi(z_1). \tag{20}$$

The latent update combines the linear predictions $K_k z_0$ according to the mixture weights determined by the initial state. For multi-step prediction over a horizon H, we apply the same mixture weights at every time step, yielding the IC-gating update rule. Using mixture weights fixed by the initial condition ensures that the same convex combination of Koopman operators is applied at every time step, preserving linear evolution in the latent space and enabling consistent multi-step prediction over long horizons.

V. Training Methodology

A. Multi-Step Linearity Loss

A core principle in our training is enforcing Koopman linearity across the entire training horizon. Specifically, we consider dense multi-step horizons $\mathcal{H}_{\text{train}} = \{1, 2, 3, \ldots, 100\}$, so that the learned latent representation evolves linearly at each timestep. For each horizon $h \in \mathcal{H}_{\text{train}}$, the latent linearity constraint reads

$$z_h = K^h z_0 \quad \Leftrightarrow \quad \phi(x_h) \approx K^h \phi(x_0). \tag{21}$$

Dense supervision ensures that the Koopman operator captures not only local, one-step transitions but also the compounding effect of multi-step evolution. Without this, errors can accumulate quickly during rollout, leading to trajectory divergence even if one-step predictions appear accurate. For MoE models, this becomes a weighted linear combination over experts:

$$z_{t+1} = \sum_{k=1}^{M} \pi_k(z_0) K_k z_t, \quad \mathcal{L}_{\text{lin}}^{\text{MoE}} = \frac{1}{|\mathcal{H}_{\text{train}}|} \sum_{h=1}^{100} \sum_{k=1}^{M} \pi_k(z_0) \| z_h^{\text{true}} - K_k^h z_0 \|_2^2. \tag{22}$$

Using the initial condition to fix mixture weights ensures that the latent dynamics remain linear for the entire rollout, allowing repeated application of the same convex combination of Koopman operators.

B. Loss Function Components

We combine multiple other loss components to jointly enforce reconstruction fidelity, latent linearity, operator stability, and expert utilization:

$$\mathcal{L} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{pred}} \mathcal{L}_{\text{pred}} + \lambda_{\text{lin}} \mathcal{L}_{\text{lin}} + \lambda_{\text{bi}} \mathcal{L}_{\text{bi}} + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{sparse}} \mathcal{L}_{\text{sparse}}. \tag{23}$$

1) Reconstruction Loss: The reconstruction loss ensures the encoder-decoder pair forms a faithful autoencoder. Minimizing this term encourages the latent space to retain sufficient information to reconstruct the original states.

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^{N} \| x_i - \psi(\phi(x_i)) \|_2^2. \tag{24}$$

2) Prediction Loss: The one-step prediction loss ensures that each operator produces accurate immediate transitions. This term provides a baseline constraint that guides the operators before multi-step linearity is enforced.

$$\mathcal{L}_{\text{pred}} = \frac{1}{N} \sum_{i=1}^{N} \| x_{i,1} - \hat{x}_{i,1} \|_2^2. \tag{25}$$

3) Multi-Step Linearity Loss: The multi-step linearity loss penalizes deviation from linear evolution across the training horizon, with $z_{i,h}^{\text{true}} = \phi(x_{i,h})$ and $z_{i,h}^{\text{pred}} = K^h z_{i,0}$. This loss explicitly constrains the latent trajectory to remain linear under repeated application of the learned operators, which is critical for stable long-horizon predictions.

$$\mathcal{L}_{\text{lin}} = \frac{1}{|\mathcal{H}_{\text{train}}|} \sum_{h \in \mathcal{H}_{\text{train}}} \frac{1}{N} \sum_{i=1}^{N} \| z_{i,h}^{\text{true}} - z_{i,h}^{\text{pred}} \|_2^2, \tag{26}$$

4) Bidirectional Consistency Loss: To ensure that forward and backward Koopman operators approximate inverses. This term encourages the latent space dynamics to be invertible, improving robustness and aiding in regularizing the learned operators.

$$\mathcal{L}_{\text{bi}} = \sum_{k=1}^{M} \|K_k K_k^{-1} - I\|_F^2. \tag{27}$$

5) Spectral Radius Penalty: We constrain the eigenvalues of each K_k to remain close to the unit circle. This preserves approximate contractivity in the latent dynamics and prevents blow-up during long-horizon rollout, with $\rho_{\text{lower}} = 0.995$ and $\rho_{\text{upper}} = 1.005$.

$$\mathcal{L}_{\text{spec}} = \sum_{k=1}^{M} \left[\max(0, \rho(K_k) - \rho_{\text{upper}})^2 + \max(0, \rho_{\text{lower}} - \rho(K_k))^2\right], \tag{28}$$

6) Load Balancing Loss (MoE only): To prevent expert collapse, we encourage uniform utilization:

$$\mathcal{L}_{\text{bal}} = \sum_{k=1}^{M} \left(\bar{\pi}_k - \frac{1}{M}\right)^2, \quad \bar{\pi}_k = \frac{1}{N}\sum_{i=1}^{N} \pi_k(z_{i,0}). \tag{29}$$

7) Sparsity Regularization: We apply L1 regularization on all network parameters which encourages compact representations and reduces overfitting.

$$\mathcal{L}_{\text{sparse}} = \sum_{\theta \in \Theta} |\theta|, \tag{30}$$

C. Loss Weights

After hyperparameter tuning, we set $\lambda_{\text{rec}} = 2.0, \lambda_{\text{pred}} = 2.0, \lambda_{\text{lin}} = 12.0, \lambda_{\text{bi}} = 1.0, \lambda_{\text{spec}} = 5.0, \lambda_{\text{bal}} = 1.0, \lambda_{\text{sparse}} = 10^{-4}$ The large weight on $\mathcal{L}_{\text{lin}}$ emphasizes the importance of multi-step linearity for accurate long-horizon prediction.

D. Precomputed Latents for Efficiency

To reduce computation during dense multi-step supervision, we precompute latent encodings at each epoch. This caching avoids repeated encoder forward passes, which dominate the computational cost of $\mathcal{L}_{\text{lin}}$, yielding approximately a $10\times$ speedup.

$$\mathcal{Z} = \{z_h = \phi(x_h) : h \in \{0\} \cup \mathcal{H}_{\text{train}}\}. \tag{31}$$

E. Training Configuration

We train using Adam with learning rate 10^{-3}, batch size 2048, and a maximum of 500 epochs with early stopping (patience = 25). A separate validation split of trajectories is used for model selection. These settings balance computational efficiency with convergence stability while providing sufficient data for accurate multi-step supervision.

VI. Experimental Setup

A. Dynamical Systems

We evaluate models on four canonical nonlinear systems that span a range of complexity, from low-dimensional oscillators to chaotic high-dimensional systems. This selection allows us to test both interpolation and extrapolation performance under diverse dynamics. Table X summarizes key system parameters.

1) Duffing Oscillator (2D): The Duffing oscillator is a bistable system with a double-well potential:

$$\ddot{x} + \delta\dot{x} - \alpha x + \beta x^3 = \gamma\cos(\omega t), \tag{32}$$

represented in state-space form as $\mathbf{x} = [x, \dot{x}]^\top$. We set $\delta = 0.3$, $\alpha = 1.0$, $\beta = 1.0$, $\gamma = 0.5$, $\omega = 1.2$. This system tests the ability of models to handle multiple attractors.

2) Van der Pol Oscillator (2D): A self-excited oscillator exhibiting a stable limit cycle:

$$\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0, \tag{33}$$

with $\mu = 1.0$. This system evaluates how well models learn oscillatory nonlinear dynamics with bounded attractors.

3) Lorenz Attractor (3D): The classical chaotic system:

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z, \tag{34}$$

with parameters $\sigma = 10$, $\rho = 28$, $\beta = 8/3$. The largest Lyapunov exponent is $\lambda_{\text{Lyap}} \approx 0.9$, indicating a predictability horizon of roughly 50–100 steps at $\Delta t = 0.01$. This system tests model performance under sensitive dependence on initial conditions.

4) Double Pendulum (4D): A mechanical chaotic system with state $\mathbf{x} = [\theta_1, \theta_2, \omega_1, \omega_2]^\top$ representing joint angles and angular velocities. Equations of motion are derived from the Lagrangian with unit masses and lengths. This system evaluates high-dimensional, strongly nonlinear interactions.

B. Data Generation

For each system, we generate 100000 trajectories using fourth-order Runge-Kutta integration with fixed timestep $\Delta t = 0.01$ s and total simulation time $T = 10$ s (1001 timesteps per trajectory). Initial conditions are sampled uniformly across system-specific ranges to cover multiple attractor basins. This ensures broad coverage of the state space for training, validation, and testing. We perform a 80-10-10 trajectory-level split to avoid data leakage. No trajectory appears in more than one set, guaranteeing that validation and test metrics measure generalization to entirely unseen trajectories. For each trajectory, we extract the initial state x_0 and states at all training horizons $\{x_1, x_2, \ldots, x_{100}\}$ This produces training samples with 101 associated states each, for dense multi-step linearity supervision.

C. Models Compared

We implement and compare seven models spanning classical methods, neural Koopman operators, and our proposed mixture architecture. Table XI summarizes parameter counts.

1) VAR (Vector Autoregression): A purely linear baseline. For multivariate time series $\mathbf{x}_t \in \mathbb{R}^{n_x}$, a VAR($p$) model evolves as below, and where $A_i \in \mathbb{R}^{n_x \times n_x}$ are coefficient matrices and ε_t is white noise. The lag order p is selected automatically via the Akaike Information Criterion (AIC) over candidates $p \in \{1, \ldots, 10\}$. VAR provides a performance floor - Any nonlinear model should exceed this baseline.

$$\mathbf{x}_t = \sum_{i=1}^{p} A_i \mathbf{x}_{t-i} + \varepsilon_t \tag{35}$$

2) eDMD (Extended Dynamic Mode Decomposition): A classical Koopman method using fixed dictionary functions $\boldsymbol{\phi}(\mathbf{x})$. Following [4], we construct a dictionary of complete polynomial basis up to degree 5 (all monomials $x_1^{p_1} x_2^{p_2} \cdots x_{n_x}^{p_{n_x}}$ with $\sum_i p_i \leq 5$) and trigonometric terms $(\sin(x_i), \cos(x_i)$ for each state variable). The Koopman operator K is computed via regularized least squares where X, X' are consecutive state pairs and $\lambda = 10^{-6}$ is the regularization parameter. Predictions evolve linearly in the lifted space: $\boldsymbol{\phi}(\mathbf{x}_{t+1}) \approx K\boldsymbol{\phi}(\mathbf{x}_t)$.

$$K = \boldsymbol{\phi}(X')^\top \boldsymbol{\phi}(X) \left(\boldsymbol{\phi}(X)^\top \boldsymbol{\phi}(X) + \lambda I\right)^{-1} \tag{36}$$

3) KAE Baseline (Koopman Autoencoder): A neural Koopman autoencoder trained with simple 1-step objectives. The architecture consists of Encoder $\phi : \mathbb{R}^{n_x} \to \mathbb{R}^{n_z}$: 4-layer MLP with LayerNorm and ReLU (hidden dim 128), Decoder $\psi : \mathbb{R}^{n_z} \to \mathbb{R}^{n_x}$: symmetric 4-layer MLP, and Koopman operator $A_f \in \mathbb{R}^{n_z \times n_z}$: learnable matrix, initialized near identity. Dynamics evolve linearly in latent space: $z_{t+1} = A_f z_t$. Training minimizes the loss below:

$$\mathcal{L}_{\text{kae}} = \underbrace{\|\mathbf{x} - \psi(\phi(\mathbf{x}))\|^2}_{\text{reconstruction}} + \underbrace{\|\mathbf{x}_{t+1} - \psi(A_f \phi(\mathbf{x}_t))\|^2}_{\text{1-step prediction}} + \underbrace{\|A_f \phi(\mathbf{x}_t) - \phi(\mathbf{x}_{t+1})\|^2}_{\text{linearity}} \tag{37}$$

4) Advanced KAE: The same architecture as KAE Baseline, but trained with the full multi-step loss function (Eq. 23) including dense multistep linearity at horizons $\mathcal{H}_{\text{train}} = \{1, 2, \ldots, 100\}$, bidirectional consistency: $A_f A_b \approx I$, spectral radius regularization: $\rho(A_f) \approx 1$, and sparsity on encoder/decoder weights.

5) MoKE (Mixture of Koopman Experts): Our proposed architecture with $M \in \{2, 3, 4\}$ Koopman operators

D. Evaluation Protocol

Models are evaluated at 25 prediction horizons covering in-distribution, near-extrapolation, and far-extrapolation regimes. To reduce computation, we evaluate a representative subset of 100 test trajectories per system. Horizons within the training range test interpolation $\mathcal{H}_{\text{eval}}^{\text{in}} = \{1, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$. This allows us to observe error accumulation within the trained range. Horizons just beyond training assess short-term generalization $\mathcal{H}_{\text{eval}}^{\text{near}} = \{110, 120, \ldots, 200\}$. For chaotic systems, these horizons correspond to roughly 1-2 Lyapunov times beyond the training horizon, capturing the onset of sensitivity to initial conditions. Long-horizon evaluation tests stability and attractor preservation $\mathcal{H}_{\text{eval}}^{\text{far}} = \{300, 400, 500, 1000\}$. This range is $3\times$–$10\times$ the training horizon, critical for assessing whether learned dynamics produce bounded trajectories over extended rollout.

E. Evaluation Metrics

We use multiple metrics to capture different aspects of predictive performance:

1) One-Step MSE: Measures immediate prediction error:

$$\text{MSE}_1 = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \|x_{i,1} - \hat{x}_{i,1}\|_2^2. \tag{38}$$

2) Multi-Step NRMSE: Normalized error at horizon H:

$$\text{NRMSE}_H = \sqrt{\frac{1}{N_{\text{test}} \cdot n_x} \sum_{i=1}^{N_{\text{test}}} \sum_{d=1}^{n_x} \left(\frac{x_{i,H}^{(d)} - \hat{x}_{i,H}^{(d)}}{\sigma_d} \right)^2}, \tag{39}$$

where σ_d is the standard deviation of dimension d in the test set. NRMSE = 1 indicates prediction is no better than the mean.

3) Chamfer Distance: Quantifies similarity between predicted and true phase-space trajectories:

$$d_{\text{Chamfer}}(\mathcal{T}, \hat{\mathcal{T}}) = \frac{1}{|\mathcal{T}|} \sum_{x \in \mathcal{T}} \min_{\hat{x} \in \hat{\mathcal{T}}} \|x - \hat{x}\|_2 + \frac{1}{|\hat{\mathcal{T}}|} \sum_{\hat{x} \in \hat{\mathcal{T}}} \min_{x \in \mathcal{T}} \|x - \hat{x}\|_2. \tag{40}$$

4) Divergence Rate: The divergence rate quantifies the exponential growth of prediction errors over time, providing a measure of model stability during rollout. For each trajectory, we first compute the prediction error at timestep t:

$$\epsilon_t = x_t - \hat{x}_t. \tag{41}$$

We then fit a linear model to the logarithm of the absolute errors to estimate the growth rate:

$$\log(|\epsilon_t|) = a \cdot t + b, \tag{42}$$

where a represents the divergence rate. Fitting is performed separately for each state dimension d:

$$a_d = \arg\min_{a,b} \sum_{t=1}^{T} \left(\log\left(|\epsilon_t^{(d)}| + \delta\right) - (a \cdot t + b) \right)^2, \tag{43}$$

with a small constant $\delta = 10^{-8}$ added for numerical stability. The overall divergence rate for a trajectory is computed as the root-mean-square across all n_x dimensions:

$$a_{\text{div}} = \sqrt{\frac{1}{n_x} \sum_{d=1}^{n_x} a_d^2}. \tag{44}$$

Additionally, a trajectory is flagged as *diverged* if any predicted state becomes numerically infinite ($|\hat{x}_t| \to \infty$). This metric is crucial for evaluating long-horizon stability, particularly in chaotic or stiff dynamical systems.

5) Spectral Radius: Used for operator stability analysis:

$$\rho(K) = \max_i |\lambda_i(K)|, \tag{45}$$

computed via power iteration. Operators with $\rho(K) \leq 1$ produce bounded latent trajectories over long rollouts.

VII. Results

We present comprehensive experimental results evaluating the Mixture of Koopman Experts (MoE) framework against baseline methods across four canonical dynamical systems. All metrics are computed over 100 held-out test trajectories per system, totaling 400 trajectories. Complete numerical results appear in Appendix Tables I–IX, with NRMSE progression curves shown in Figures 2–5.

A. One-Step Prediction Accuracy

One-step prediction accuracy, reported in Table I, reveals a striking pattern: eDMD with polynomial observables achieves the lowest immediate prediction error across all four systems, with MSE values ranging from 9.32×10^{-9} on Van der Pol to 7.16×10^{-4} on Double Pendulum. This reflects the well-known strength of polynomial basis functions for capturing local dynamics when the dictionary aligns with the underlying system structure. However, as we will see, this one-step advantage does not translate to superior long-horizon performance on complex systems. Among neural methods, the baseline KAE achieves competitive one-step accuracy on simpler systems (Duffing: 1.88×10^{-5}, Van der Pol: 1.48×10^{-4}), but struggles on chaotic systems where the Lorenz one-step MSE reaches 0.13. The MoE variants show intermediate one-step performance, with MoE-4 achieving 1.53×10^{-4} on Duffing and 5.23×10^{-3} on Double Pendulum. Notably, VAR performs poorly across all systems, confirming that purely linear autoregressive models cannot capture the nonlinear dynamics present in these benchmarks.

B. Multi-Step NRMSE Progression

The multi-step NRMSE results, detailed in Tables VI–IX and summarized at H=100 in Table V, reveal the core advantage of the MoE architecture: robust performance on multi-regime systems where single-operator methods fail.

On the Duffing oscillator, MoE-4 achieves NRMSE@100 = 0.04, matching eDMD's accuracy on non-diverged trajectories while representing a 44% improvement over Advanced KAE (0.08). The advantage becomes more pronounced at extrapolation horizons: at H=200, MoE-4 maintains NRMSE = 0.09 compared to KAE's 0.51. This improvement stems from the mixture architecture's ability to learn separate Koopman operators for the two potential wells of the Duffing system, enabling smooth transitions through the saddle point that separates them.

On the Van der Pol oscillator, eDMD demonstrates exceptional performance with NRMSE remaining below 0.01 through H=200 and reaching only 0.04 at H=1000. This near-perfect tracking reflects the excellent match between polynomial observables and limit-cycle dynamics. The MoE variants achieve NRMSE@100 between 0.03–0.05, competitive with Advanced KAE but unable to match eDMD's polynomial precision on this single-attractor system. This result confirms our hypothesis: mixture architectures provide greatest benefit on multi-regime systems, not simple limit cycles.

On the Lorenz attractor, all models face the fundamental challenge of chaotic sensitivity. MoE-3 achieves the best in-distribution performance with NRMSE@100 = 0.36, substantially outperforming KAE (1.53) and VAR (1.24). Interestingly, eDMD shows strong performance (NRMSE@100 = 0.05) within the training horizon, but this advantage diminishes at longer horizons as all models converge toward NRMSE $\approx$ 1.0–1.5 at H=500–1000. This convergence reflects the Lyapunov predictability limit: the Lorenz system's positive Lyapunov exponent ($\lambda \approx 0.9$) causes exponential error growth until predictions become statistically indistinguishable from noise.

On the Double Pendulum, MoE-2 and MoE-4 both achieve NRMSE@100 = 0.39, representing an 11% improvement over Advanced KAE (0.44) and outperforming KAE (0.65) and VAR (0.66). At near-extrapolation horizons (H=110–200), MoE variants maintain their advantage with NRMSE values of 0.42–0.66 compared to KAE's 0.72–1.13. The four-dimensional phase space of the double pendulum contains multiple energy regimes where the mixture architecture's expert specialization proves beneficial.

C. Numerical Stability and Trajectory Divergence

Perhaps the most critical finding concerns numerical stability, summarized in Table II. Across all 400 test trajectories spanning four dynamical systems, MoE variants maintain zero divergence - no predicted trajectory becomes numerically unbounded. In stark contrast, eDMD diverges on 16% of Duffing trajectories and 15% of Double Pendulum trajectories, totaling 31 diverged trajectories (7.75% overall). This stability difference stems directly from spectral radius regularization. Table III shows that all MoE spectral radii lie within 1.00–1.01, ensuring that repeated application of the learned Koopman operators produces bounded trajectories. When eDMD fails, it fails catastrophically: the Chamfer distance becomes infinite (Table IV), rendering predictions useless for downstream applications. For safety-critical systems where occasional blow-ups are unacceptable, MoE's stability guarantee provides essential reliability that compensates for any marginal accuracy differences on stable trajectories. The practical implication is clear: while eDMD may achieve lower average error on trajectories where it succeeds, MoE provides predictable, bounded predictions across all operating conditions. This trade-off favors MoE in control, planning, and any application requiring guaranteed numerical stability.

D. Trajectory Shape Preservation

The Chamfer distance results in Table IV measure geometric similarity between predicted and true trajectories in phase space, independent of temporal alignment. On Van der Pol, eDMD achieves the lowest Chamfer distance (0.016), reflecting its excellent limit-cycle tracking. On Lorenz, eDMD again performs best (3.315) among models with finite distances. However, on Duffing and Double Pendulum, eDMD's divergence produces infinite Chamfer distances, while MoE variants achieve finite values (Duffing: 0.51–0.54, Double Pendulum: 10.1–10.2). Advanced KAE achieves the best Chamfer distance on Duffing (0.267), suggesting that for attractor-preserving predictions, the advanced training objectives contribute significantly even without mixture decomposition.

E. Summary

The experimental results validate the core hypothesis motivating the MoE architecture. On multi-regime systems - the bistable Duffing oscillator and the high-dimensional Double Pendulum - MoE achieves 11–44% NRMSE improvement over single-operator Koopman methods while maintaining perfect numerical stability. On the simple Van der Pol limit cycle, classical eDMD with polynomial observables remains superior, demonstrating that mixture complexity is unnecessary when the system admits a single global linearization. On chaotic systems, all models converge to similar long-horizon error levels, reflecting fundamental Lyapunov predictability limits rather than model deficiencies. The zero-divergence property of MoE across all 400 test trajectories, compared to 7.75% divergence for eDMD, establishes the framework as a reliable choice for applications requiring guaranteed bounded predictions.

VIII. Discussion

Our experiments reveal that MoE provides meaningful advantages on dynamical systems with multiple regimes or complex state-space structure, while offering limited benefit on simpler single-attractor dynamics. On the Duffing oscillator, where trajectories navigate between two potential wells separated by a saddle point, MoE-4 achieves NRMSE@100 = 0.04, matching eDMD's accuracy on non-diverged trajectories while maintaining bounded predictions where eDMD catastrophically diverges on 16% of trajectories. Similarly, on the four-dimensional Double Pendulum, MoE-2 and MoE-4 achieve NRMSE@100 = 0.39, an 11% improvement over Advanced KAE (0.44), while eDMD diverges on 15% of trajectories. These results validate the core hypothesis: multiple locally-linear Koopman operators, each specializing to distinct dynamical regimes, can collectively approximate global nonlinear dynamics more faithfully than a single global linearization. However, on the Van der Pol oscillator - a system with a single stable limit cycle - eDMD with polynomial observables achieves near-perfect long-horizon tracking (NRMSE@1000 = 0.04), demonstrating that when classical observables naturally match the system structure, the additional complexity of a mixture architecture provides no benefit.

The most critical finding concerns numerical stability. Across all 400 test trajectories spanning four dynamical systems, MoE variants maintain zero divergence, while eDMD diverges on 31 trajectories (7.75%). This reliability advantage stems directly from spectral radius regularization, which constrains learned Koopman operators to have eigenvalues near the unit circle (all MoE spectral radii are within 1.00–1.01 as shown in Table III). When eDMD fails, it fails catastrophically - producing infinite Chamfer distances and rendering predictions useless for downstream tasks. In contrast, MoE trades marginally higher average error for guaranteed bounded predictions, a property essential for control, planning, and any application that cannot tolerate occasional numerical blow-ups. The practical implication is clear: for safety-critical or reliability-sensitive applications, MoE's stability guarantees may outweigh eDMD's superior average-case performance on stable trajectories.

On chaotic systems (Lorenz and Double Pendulum), all models converge to similar error levels at long horizons, with NRMSE approaching 1.0–2.0 regardless of model architecture. This convergence reflects a fundamental physical limit rather than a modeling deficiency: chaotic systems exhibit exponential sensitivity to initial conditions characterized by positive Lyapunov exponents, meaning that prediction errors grow exponentially until they saturate at the scale of the attractor. No data-driven method can circumvent this intrinsic unpredictability - the predictability horizon is set by the physics, not the algorithm. Within the discriminative window ($H \leq 100$–200 depending on system), model differences remain meaningful, and MoE variants consistently achieve competitive or best performance. Beyond this window, model selection becomes irrelevant as all predictions degrade to statistical baseline levels.

IX. Limitations and Future Work

Several limitations constrain the scope of our findings and suggest directions for future research. First, our hyperparameter selection was limited by computational resources; with additional compute and more extensive data processing pipelines, systematic hyperparameter optimization via grid search or Bayesian methods could yield substantially improved performance across all models, potentially altering the relative rankings we report. Second, our evaluation is restricted to four well-studied low-dimensional systems (2–4 state dimensions), and the scalability of MoE to high-dimensional systems such as fluid dynamics or molecular simulations remains unexplored. Third, the current architecture assumes full state observability - extending MoE to partial observations via delay embeddings or learned observation models is a natural next step. Fourth, while we demonstrate that experts specialize to different state-space regions, we lack theoretical guarantees on convergence or optimality of this specialization. Finally, integrating MoE with model-predictive control to exploit its stability guarantees in closed-loop settings represents a promising avenue for future investigation.

X. Conclusion

We have presented Mixture of Koopman Experts (MoE), a neural architecture that extends Koopman operator learning through state-dependent combination of multiple locally-linear dynamics. Comprehensive experiments across four dynamical systems - Duffing oscillator, Van der Pol oscillator, Lorenz attractor, and double pendulum - demonstrate that MoE achieves competitive prediction accuracy on multi-regime systems (NRMSE@100 = 0.04 on Duffing, 0.39 on Double Pendulum) while maintaining perfect numerical stability (0% divergence across 400 test trajectories, compared to 7.75% for eDMD). The architecture's advantage is most pronounced on systems with multiple equilibria or complex state-space structure, where learned experts naturally specialize to distinct dynamical regimes; conversely, on simple single-attractor systems like the Van der Pol limit cycle, classical eDMD with polynomial observables remains superior (NRMSE@1000 = 0.04). Our analysis also reveals fundamental predictability limits: on chaotic systems, all models converge to similar error levels beyond the Lyapunov time, establishing that sophisticated architectures provide returns only within the system's inherent predictability horizon. Future work will explore adaptive expert allocation, integration with model-predictive control, and extension to partial observations and higher-dimensional systems where the mixture-of-experts paradigm may offer even greater benefits.

A. Code Availability

All code, data generation scripts, and trained models are available at the link below (vm_linux_main is the branch with the final code and results):
https://github.com/pujiacheng/ESE5380_MoKE-on-non-linear-dynamical-system/tree/vm_linux_main

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APPENDIX

TABLE I: One-Step MSE across all systems. Best per column in **bold**.

Model	Duffing	Van der Pol	Lorenz	Double Pend.
VAR	2.60e-04	1.49e-03	2.05	3.60e-03
eDMD	**5.41e-07**	**9.32e-09**	**1.16e-04**	**7.16e-04**
KAE	1.88e-05	1.48e-04	0.13	2.24e-03
Adv. KAE	4.11e-04	1.18e-03	0.05	4.08e-03
MoE-2	9.75e-04	2.29e-04	0.06	4.53e-03
MoE-3	3.93e-04	3.25e-04	0.06	8.13e-03
MoE-4	1.53e-04	2.00e-04	0.08	5.23e-03

TABLE II: Divergence rate (% of 100 test trajectories that diverged).

Model	Duffing	Van der Pol	Lorenz	Double Pend.
VAR	0%	0%	0%	0%
eDMD	16%	0%	0%	15%
KAE	0%	0%	0%	0%
Adv. KAE	0%	0%	0%	0%
MoE-2	0%	0%	0%	0%
MoE-3	0%	0%	0%	0%
MoE-4	0%	0%	0%	0%

TABLE III: Spectral radius $\rho(K)$ of learned Koopman operators.

Model	Duffing	Van der Pol	Lorenz	Double Pend.
KAE	—	—	—	—
Adv. KAE	0.9999	1.0000	1.0000	1.0066
MoE-2	1.0053	1.0025	0.9999	1.0031
MoE-3	1.0005	1.0033	1.0068	1.0068
MoE-4	1.0039	1.0051	1.0025	1.0120

TABLE IV: Chamfer distance at H=1000. ∞ = diverged.

Model	Duffing	Van der Pol	Lorenz	Double Pend.
VAR	2.141	2.650	13.207	11.755
eDMD	∞	0.016	3.315	∞
KAE	0.945	2.515	16.259	10.581
Adv. KAE	0.267	0.432	9.930	10.664
MoE-2	0.516	1.259	8.594	10.123
MoE-3	0.536	0.977	9.059	10.126
MoE-4	0.508	0.835	9.387	10.247

TABLE V: NRMSE at H=100. Best per column in **bold**.

Model	Duffing	Van der Pol	Lorenz	Double Pend.
VAR	0.72	0.47	1.24	0.66
eDMD	0.04	**0.00**	**0.05**	0.47
KAE	0.18	0.24	1.53	0.65
Adv. KAE	0.08	0.05	0.66	0.44
MoE-2	0.06	0.05	0.57	**0.39**
MoE-3	0.06	0.05	0.36	0.41
MoE-4	**0.04**	0.03	0.57	0.39

TABLE VI: Duffing: NRMSE at all horizons. — = diverged/inf.

H	VAR	eDMD	KAE	Adv.KAE	MoE-2	MoE-3	MoE-4
In-Distribution ($H \leq 100$)							
10	0.11	0.01	0.02	0.03	0.04	0.04	0.02
20	0.20	0.01	0.03	0.03	0.04	0.04	0.02
30	0.29	0.02	0.04	0.04	0.05	0.04	0.03
40	0.37	0.02	0.06	0.05	0.05	0.04	0.03
50	0.45	0.02	0.08	0.05	0.05	0.05	0.03
60	0.51	0.03	0.09	0.06	0.05	0.05	0.04
70	0.57	0.03	0.11	0.06	0.05	0.05	0.04
80	0.63	0.04	0.13	0.07	0.05	0.05	0.04
90	0.68	5.61	0.15	0.07	0.06	0.05	0.04
100	0.72	0.04	0.18	0.08	0.06	0.06	0.04
Near-Extrapolation ($100 < H \leq 200$)							
110	0.76	0.05	0.21	0.09	0.06	0.06	0.05
120	0.80	—	0.23	0.10	0.06	0.06	0.05
130	0.84	0.06	0.27	0.10	0.06	0.07	0.05
140	0.87	0.06	0.30	0.11	0.06	0.07	0.06
150	0.90	0.07	0.33	0.12	0.07	0.07	0.06
160	0.93	0.07	0.37	0.12	0.07	0.07	0.07
170	0.96	0.07	0.41	0.13	0.07	0.08	0.07
180	0.99	0.08	0.44	0.14	0.08	0.08	0.08
190	1.01	0.08	0.48	0.15	0.08	0.09	0.08
200	1.03	0.09	0.51	0.16	0.09	0.09	0.09
Far-Extrapolation ($H > 200$)							
300	1.18	0.15	0.86	0.27	0.28	0.18	0.22
400	1.24	0.22	1.18	0.38	0.58	0.30	0.47
500	1.29	0.30	1.40	0.47	0.84	0.48	0.76
1000	1.37	0.62	1.59	0.93	1.68	1.06	1.48

TABLE VII: Van der Pol: NRMSE at all horizons. — = diverged/inf.

H	VAR	eDMD	KAE	Adv.KAE	MoE-2	MoE-3	MoE-4
In-Distribution ($H \leq 100$)							
10	0.11	0.00	0.02	0.04	0.03	0.03	0.02
20	0.18	0.00	0.03	0.04	0.04	0.04	0.03
30	0.24	0.00	0.05	0.05	0.05	0.04	0.03
40	0.29	0.00	0.08	0.05	0.05	0.05	0.03
50	0.33	0.00	0.10	0.05	0.05	0.05	0.03
60	0.37	0.00	0.13	0.05	0.05	0.05	0.03
70	0.40	0.00	0.15	0.05	0.05	0.05	0.03
80	0.43	0.00	0.18	0.05	0.05	0.05	0.03
90	0.45	0.00	0.21	0.05	0.05	0.05	0.03
100	0.47	0.00	0.24	0.05	0.05	0.05	0.03
Near-Extrapolation ($100 < H \leq 200$)							
110	0.49	0.01	0.27	0.05	0.05	0.05	0.03
120	0.51	0.01	0.31	0.05	0.05	0.05	0.04
130	0.53	0.01	0.35	0.05	0.05	0.05	0.04
140	0.55	0.01	0.39	0.05	0.05	0.06	0.04
150	0.56	0.01	0.43	0.06	0.06	0.06	0.05
160	0.58	0.01	0.47	0.06	0.06	0.06	0.05
170	0.59	0.01	0.51	0.06	0.06	0.06	0.05
180	0.61	0.01	0.55	0.06	0.07	0.07	0.06
190	0.62	0.01	0.59	0.06	0.07	0.07	0.07
200	0.63	0.01	0.62	0.07	0.08	0.08	0.08
Far-Extrapolation ($H > 200$)							
300	0.75	0.01	0.98	0.12	0.23	0.18	0.20
400	0.84	0.01	1.27	0.20	0.49	0.33	0.37
500	0.88	0.02	1.44	0.24	0.73	0.48	0.50
1000	0.96	0.04	1.79	0.31	1.74	1.03	0.85

TABLE VIII: Lorenz: NRMSE at all horizons. — = diverged/inf.

H	VAR	eDMD	KAE	Adv.KAE	MoE-2	MoE-3	MoE-4
In-Distribution ($H \leq 100$)							
10	0.62	0.01	0.36	0.15	0.11	0.10	0.12
20	0.97	0.01	0.99	0.37	0.24	0.18	0.23
30	1.10	0.01	1.29	0.52	0.33	0.23	0.34
40	1.14	0.01	1.37	0.58	0.39	0.26	0.42
50	1.15	0.02	1.40	0.59	0.43	0.28	0.45
60	1.17	0.02	1.42	0.60	0.45	0.28	0.48
70	1.19	0.02	1.46	0.62	0.49	0.29	0.50
80	1.22	0.03	1.51	0.64	0.52	0.30	0.53
90	1.24	0.04	1.53	0.65	0.55	0.33	0.55
100	1.24	0.05	1.53	0.66	0.57	0.36	0.57
Near-Extrapolation ($100 < H \leq 200$)							
110	1.24	0.06	1.53	0.68	0.59	0.39	0.59
120	1.24	0.07	1.52	0.69	0.60	0.41	0.61
130	1.23	0.08	1.52	0.71	0.61	0.44	0.62
140	1.24	0.10	1.52	0.72	0.63	0.48	0.64
150	1.25	0.11	1.52	0.75	0.66	0.53	0.66
160	1.26	0.13	1.53	0.77	0.68	0.58	0.69
170	1.26	0.14	1.53	0.78	0.70	0.62	0.71
180	1.25	0.16	1.53	0.79	0.71	0.65	0.72
190	1.25	0.17	1.52	0.80	0.73	0.67	0.74
200	1.24	0.19	1.51	0.81	0.74	0.69	0.75
Far-Extrapolation ($H > 200$)							
300	1.26	0.45	1.52	0.97	0.91	0.91	0.92
400	1.27	0.61	1.53	1.08	1.01	1.16	1.04
500	1.26	0.73	1.53	1.15	1.06	1.42	1.11
1000	1.28	1.06	1.53	1.26	1.20	1.99	1.46

TABLE IX: Double Pendulum: NRMSE at all horizons. — = diverged/inf.

H	VAR	eDMD	KAE	Adv.KAE	MoE-2	MoE-3	MoE-4
In-Distribution ($H \leq 100$)							
10	0.08	0.03	0.03	0.05	0.05	0.05	0.05
20	0.16	0.06	0.06	0.09	0.08	0.09	0.08
30	0.23	0.10	0.11	0.14	0.12	0.12	0.12
40	0.30	0.14	0.16	0.18	0.15	0.16	0.15
50	0.37	0.19	0.22	0.22	0.19	0.20	0.18
60	0.43	—	0.30	0.26	0.22	0.23	0.22
70	0.49	0.31	0.38	0.31	0.27	0.28	0.27
80	0.55	—	0.47	0.35	0.31	0.32	0.31
90	0.61	0.42	0.56	0.39	0.34	0.36	0.35
100	0.66	0.47	0.65	0.44	0.39	0.41	0.39
Near-Extrapolation ($100 < H \leq 200$)							
110	0.69	0.50	0.72	0.47	0.42	0.44	0.42
120	0.72	0.53	0.78	0.51	0.45	0.47	0.45
130	0.74	0.56	0.84	0.54	0.48	0.51	0.48
140	0.75	0.58	0.90	0.57	0.51	0.54	0.50
150	0.77	0.60	0.94	0.60	0.53	0.56	0.53
160	0.78	0.61	0.99	0.63	0.55	0.58	0.55
170	0.79	0.80	1.03	0.66	0.57	0.60	0.58
180	0.80	0.64	1.07	0.69	0.60	0.62	0.61
190	0.81	0.65	1.10	0.73	0.62	0.64	0.64
200	0.82	0.66	1.13	0.77	0.65	0.67	0.66
Far-Extrapolation ($H > 200$)							
300	0.91	0.78	1.36	1.13	0.86	0.85	0.90
400	0.96	0.83	1.44	1.32	0.99	0.99	1.04
500	1.01	0.89	1.48	1.42	1.10	1.10	1.17
1000	1.19	1.10	1.65	1.59	1.36	1.39	1.49

TABLE X: Summary of dynamical systems used for evaluation.

System	Dimension	Type	Key Feature
Duffing Oscillator	2D	Bistable	Double-well potential
Van der Pol	2D	Limit cycle	Self-excited oscillation
Lorenz Attractor	3D	Chaotic	Strange attractor
Double Pendulum	4D	Chaotic	High sensitivity

TABLE XI: Model parameter counts across all dynamical systems.

Model	**Van der Pol (2D)**	**Duffing (2D)**	**Lorenz (3D)**	**Double Pend. (4D)**
VAR	4	40	90	160
eDMD	625	625	3,844	17,956
KAE Baseline	37,488	37,488	39,155	40,872
Advanced KAE	104,660	104,660	106,452	108,344
MoE (2 Experts)	107,710	107,710	110,072	112,634
MoE (3 Experts)	107,943	107,943	110,555	113,467
MoE (4 Experts)	108,176	108,176	111,038	114,300

TABLE XII: Comparison of Koopman and MoE Approaches for Nonlinear Dynamics

Method Category	**Key Papers**	**Strengths**	**Limitations**
Classical Koopman			
DMD	Schmid (2010)	Efficient spectral analysis	Single global operator
EDMD	Williams et al. (2015)	Dictionary-based lifting	Requires manual basis selection
Neural Koopman			
Koopman Autoencoder	Lusch et al. (2018)	Learned nonlinear observables	Single operator limitation
Deep Koopman	Li et al. (2017)	End-to-end differentiable	Global linearization assumption
Multi-Regime Methods			
Switching Koopman	Kaiser et al. (2021)	Multiple local models	Hard transitions, no smoothness
Piecewise Linear DMD	Proctor et al. (2016)	Handles bifurcations	Requires manual partitioning
MoE Architectures			
Classical MoE	Jacobs et al. (1991)	Modular specialization	Not designed for dynamics
Sparse MoE	Shazeer et al. (2017)	Scalable gating	Input-space focus
MoKE (This Work)	—	**Soft gating + local Koopman operators in shared latent space**	—

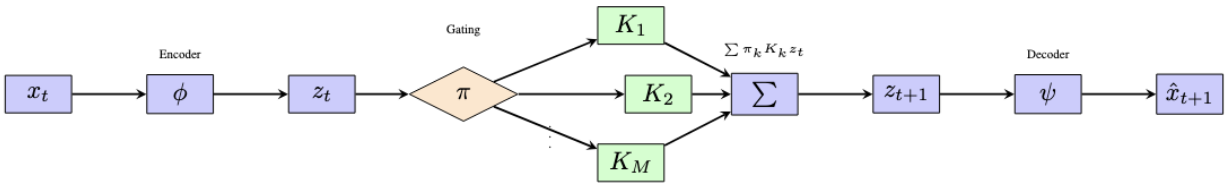


Fig. 1: MoKE architecture.

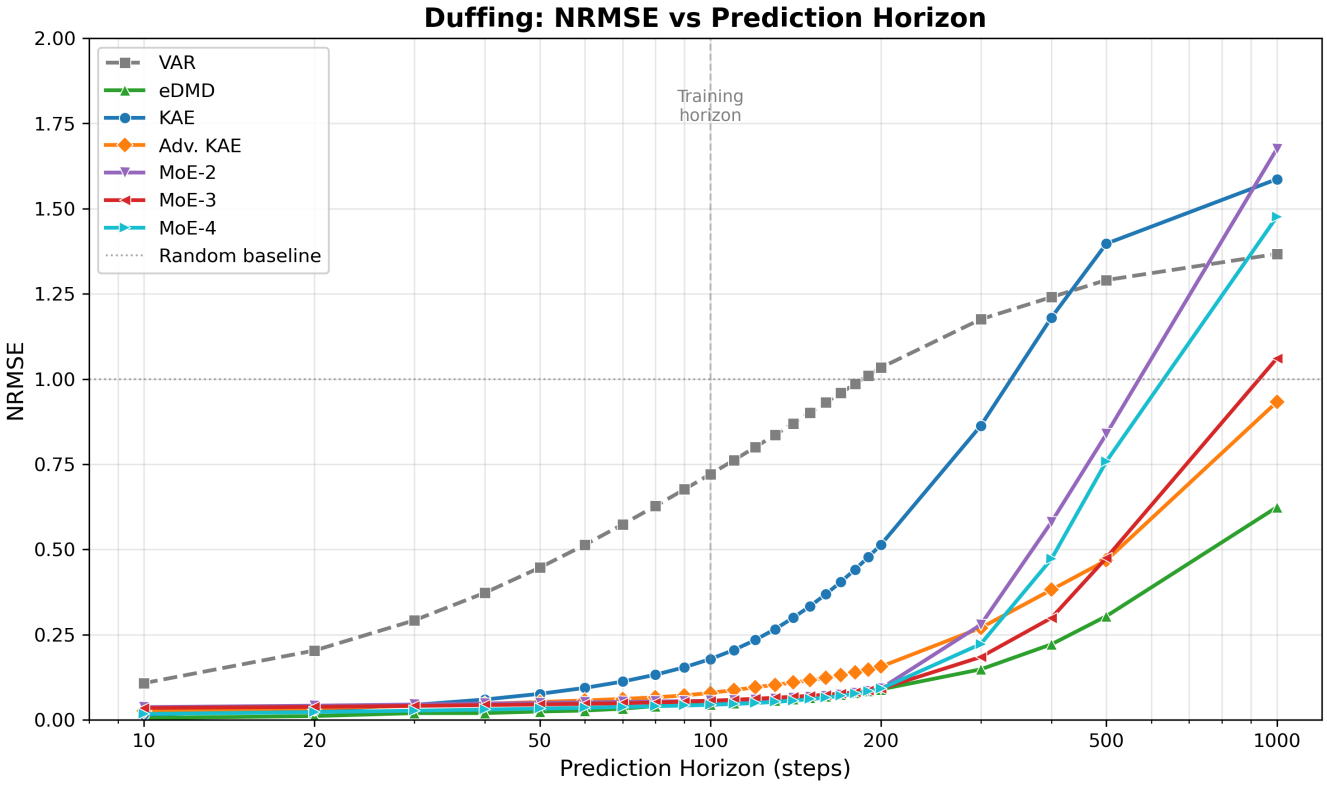


Fig. 2: Duffing: NRMSE vs. horizon.

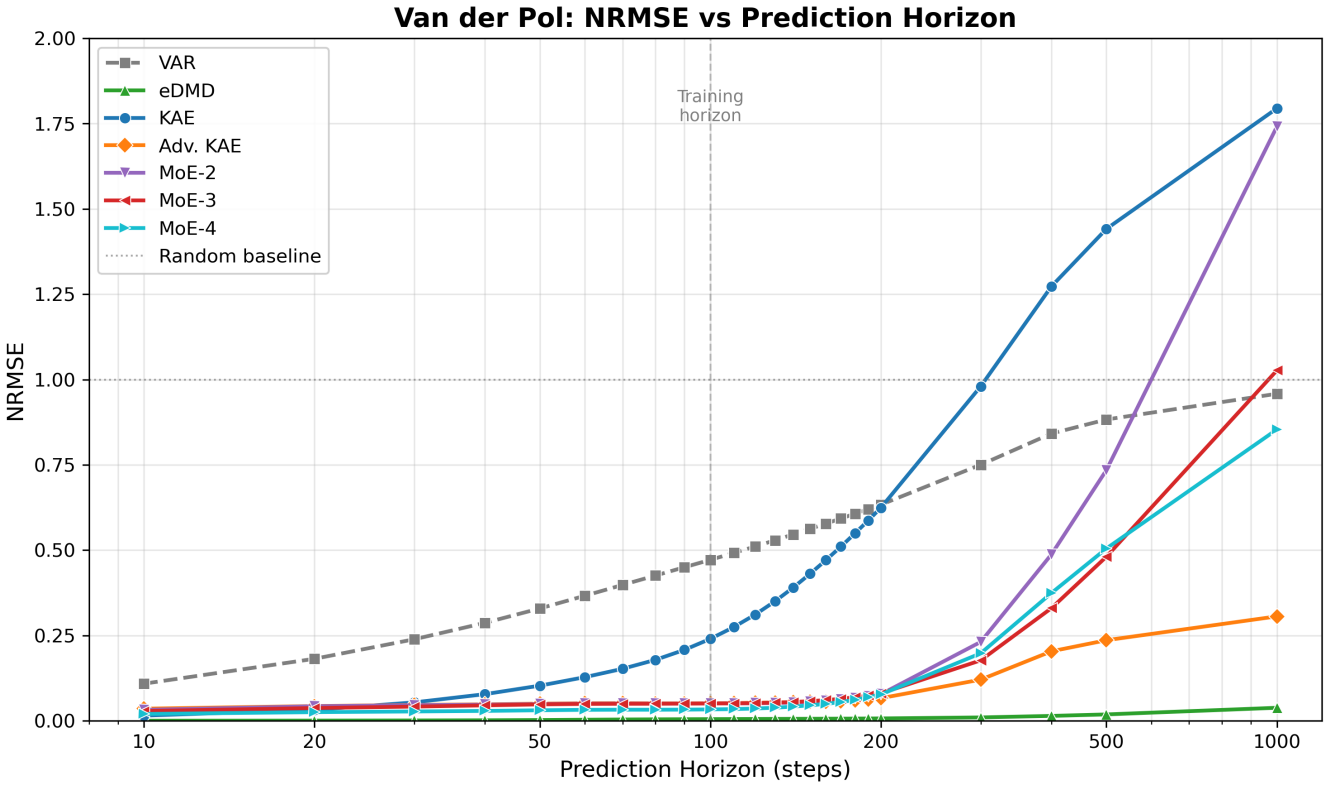


Fig. 3: Van der Pol: NRMSE vs. horizon.

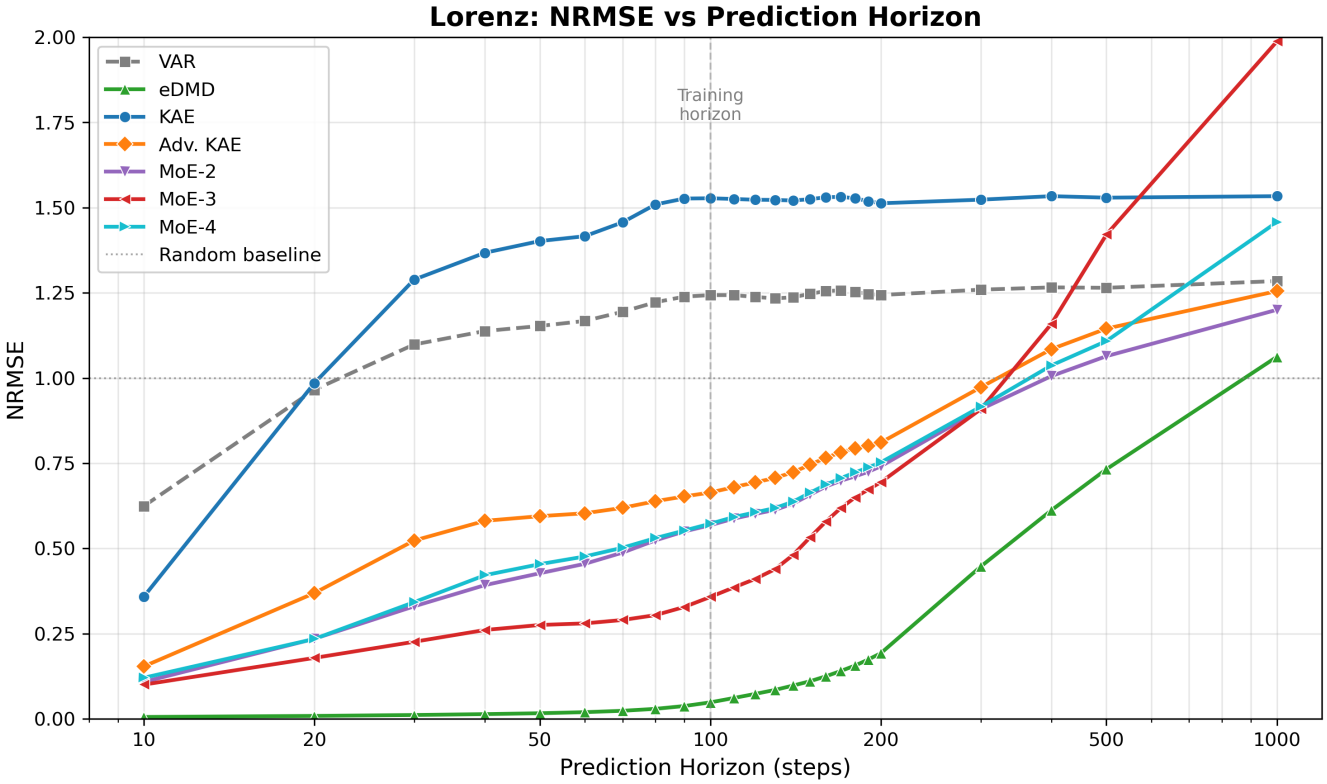


Fig. 4: Lorenz: NRMSE vs. horizon.

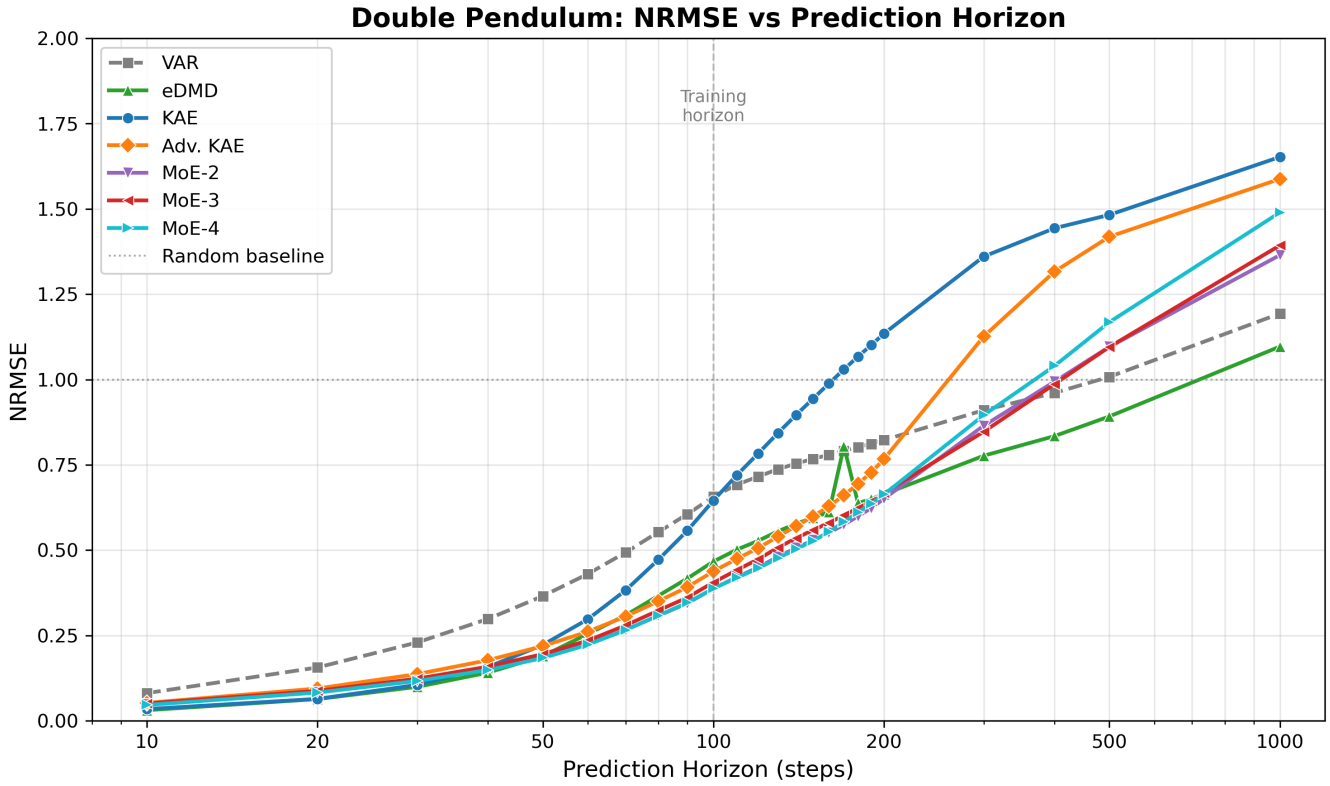


Fig. 5: Double Pendulum: NRMSE vs. horizon.

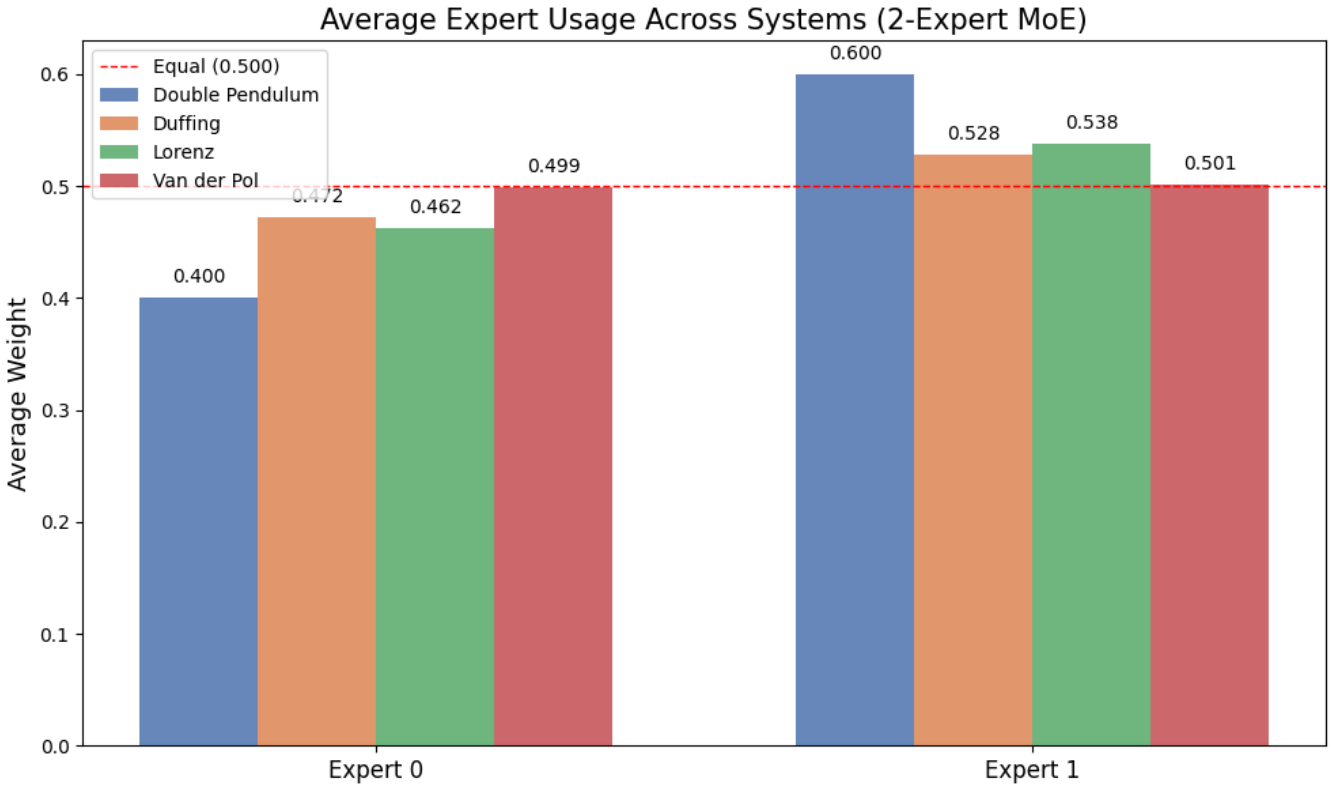


Fig. 6: Expert activation distribution for MoE-2 across all four dynamical systems. Each system shows the proportion of test samples routed to each of the two experts.

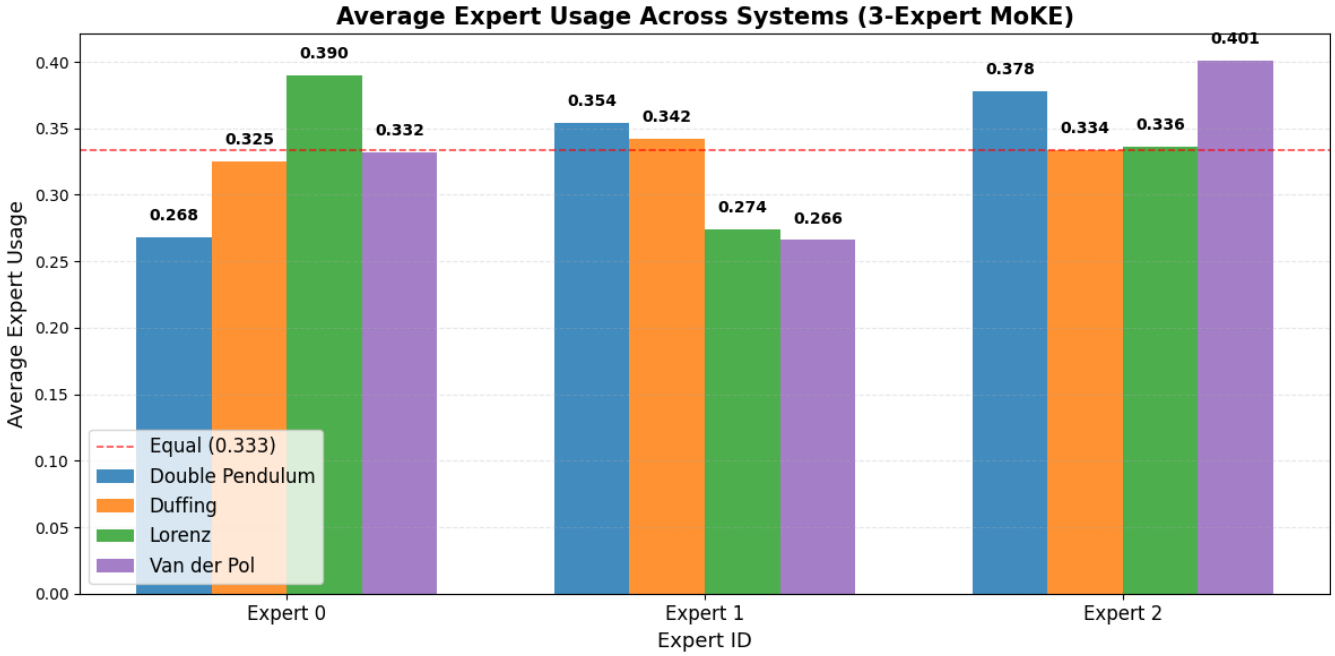


Fig. 7: Expert activation distribution for MoE-3 across all four dynamical systems. Each system shows the proportion of test samples routed to each of the three experts.

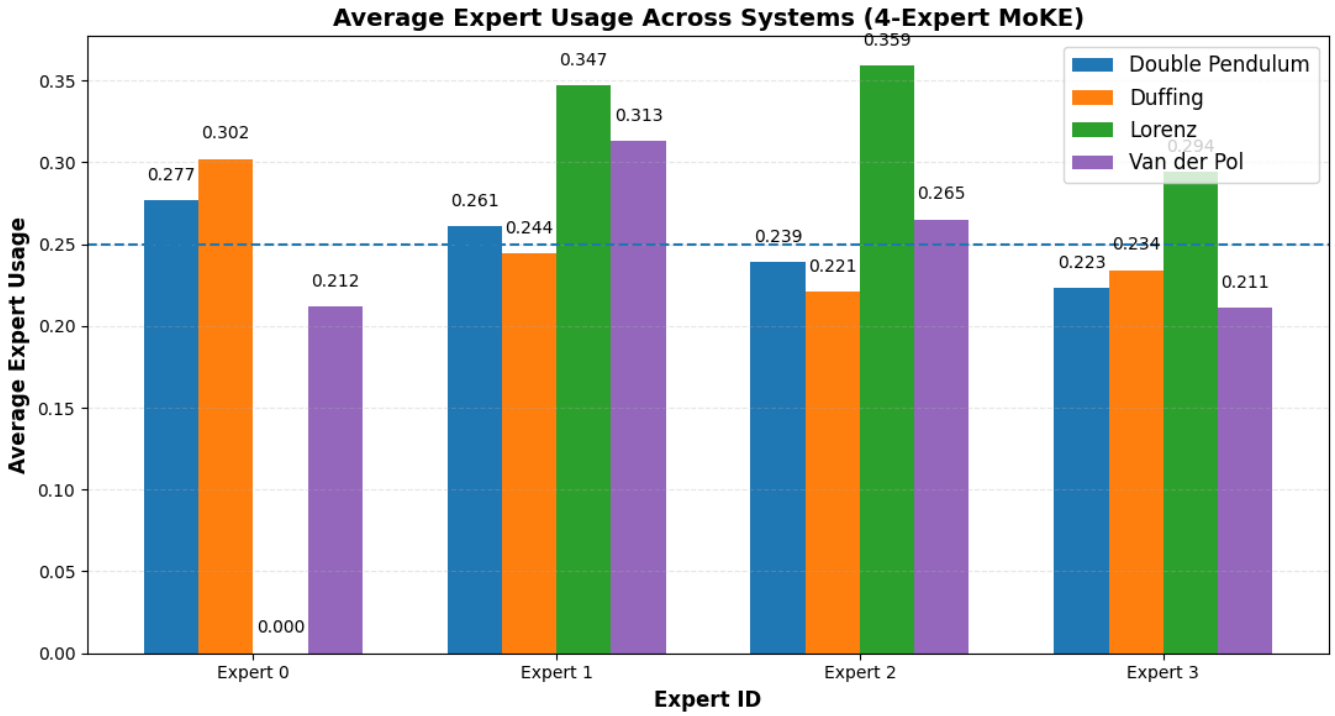


Fig. 8: Expert activation distribution for MoE-4 across all four dynamical systems. Each system shows the proportion of test samples routed to each of the four experts.

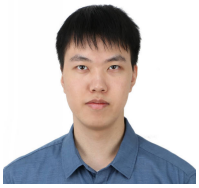
Nishant Gavhane



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